2. Quantum Computing

Sr. No.	Parameters	Brief details
1.	**Brief details of Technology Developed**	The system provides a secure way to stream live video from a drone to a ground station by using advanced, future-proof encryption methods that can withstand even quantum-computer attacks. The drone captures video, protects each frame with strong encryption, and sends it over a single connection that can pass through multiple relay drones if needed. The ground station checks the identity of the drone, safely exchanges a shared secret key, and then uses that key to unlock and display the live video. The entire setup has been tested on small, low-power devices like Raspberry Pi boards, showing that it works efficiently even on compact hardware used in field conditions.
	Name of the Project under which technology is developed	Agile and Ad hoc Free Space based Quantum Communication using Drone (Drone-QC)
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 6
	Brief Write-up on the Technology Developed	The system enables secure real-time video streaming from a drone to a ground station using advanced, quantum-resistant encryption techniques designed to remain secure against future quantum-computing threats. Video is captured and encrypted onboard the drone, transmitted over a single connection--optionally via multiple relay drones--while the ground station authenticates the drone, securely establishes shared keys, and decrypts the live feed; the solution has been validated on low-power platforms such as Raspberry Pi, demonstrating efficient operation.
	Details of Contact Person, C-DAC Centre along with Mobile No. and e-mail ID	Dr. Anindita Banerjee, C-DAC Pune 9831838821, aninditab@cdac.in
2.	**Brief details of Technology Developed**	The testbed provides a dedicated experimental platform for research, collaboration, and capacity building, enabling rapid translational research and the transition of quantum technologies from laboratory prototypes to practical deployments. It supports applications including Quantum Random Number Generation, Education and Research Modules, and Quantum Key Distribution.
	Name of the Project under which technology is developed	Agile and Ad hoc Free Space based Quantum Communication using Drone (Drone-QC)
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 5
	Brief Write-up on the Technology Developed	The testbed offers a dedicated experimental platform to support research, collaborative development, and skill development. It is designed to accelerate translational research and facilitate the transition of quantum technologies from laboratory prototypes to real-world commercial applications. **Applications:** Quantum Random Number Generation, Education and Research Modules, and Quantum Key Distribution.

Sr. No.	Parameters	Brief details
	Details of Contact Person, C-DAC Centre along with Mobile No. and e-mail ID	Dr. Anindita Banerjee, C-DAC Pune 9831838821, aninditab@cdac.in
3.	**Brief details of Technology Developed**	Free-space Quantum Key Distribution (QKD) provides an innovative solution for establishing highly secure communication links in scenarios where traditional fiber optic installations are impractical, expensive, or impossible. Its key applications lie in creating an unconditionally secure layer for critical infrastructure and future global communication networks. The solution has QKD integrated with PQC and QRNG. Application areas: Strategic and critical sectors
	Name of the Project under which technology is developed	Agile and Ad hoc Free Space based Quantum Communication using Drone (Drone-QC)
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 5
	Brief Write-up on the Technology Developed	Free-space Quantum Key Distribution (QKD) provides an innovative solution for establishing highly secure communication links in scenarios where traditional fiber optic installations are impractical, expensive, or impossible. Its key applications lie in creating an unconditionally secure layer for critical infrastructure and future global communication networks. The solution has QKD integrated with PQC and QRNG. Application areas: Strategic and critical sectors
	Details of Contact Person, C-DAC Centre along with Mobile No. and e-mail ID	Dr. Anindita Banerjee, C-DAC Pune 9831838821, aninditab@cdac.in
4.	**Brief details of Technology Developed**	Programmable Single-Qubit Photonic Quantum Computing Education toolkit involves seamless integration of a photonic-based single-qubit quantum computing system with an integrated web-based programmable software framework. This software transmits quantum gate instructions in real-time to the physical qubits and receives corresponding measurement outcomes. The toolkit helps teaching the fundamentals of quantum information science starting from quantum (q) postulates, q-gates, q-circuits, q-optics and computing. The system has been designed explicitly for quantum education and foundational industrial training. Its in-house developed software stack, encompassing front-end interfaces, middleware, and backend processing, is integrated seamlessly with quantum hardware.
	Name of the Project under which technology is developed	Agile and Ad hoc Free Space based Quantum Communication using Drone (Drone-QC)
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	TRL 5

Sr. No.	Parameters	Brief details
	Brief Write-up on the Technology Developed	Programmable Single-Qubit Photonic Quantum Computing Education toolkit involves seamless integration of a photonic-based single-qubit quantum computing system with an integrated web-based programmable software framework. This software transmits quantum gate instructions in real-time to the physical qubits and receives corresponding measurement outcomes. The toolkit helps teaching the fundamentals of quantum information science starting from quantum (q) postulates, q-gates, q-circuits, q-optics and computing. The system has been designed explicitly for quantum education and foundational industrial training. Its in-house developed software stack, encompassing front-end interfaces, middleware, and backend processing, is integrated seamlessly with quantum hardware.
	Details of Contact Person, C-DAC Centre along with Mobile No. and e-mail ID	Dr. Anindita Banerjee, C-DAC Pune 9831838821, aninditab@cdac.in
5.	**Brief details of Technology Developed**	Quantum resistant TLS/SSL Suite
	Name of the Project under which technology is developed	Development of Secure Post Quantum Public Key Infrastructure
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	4
	Brief Write-up on the Technology Developed	Transport Layer Security (TLS), is a widely adopted security protocol designed to facilitate privacy and data security for communications over the Internet. A primary use case of TLS is encrypting the communication between web applications and servers, which can be achieved by using appropriate protocols in the TLS cipher suite. The current version of TLS (version 1.3) is based on ECC (Elliptic Curve Cryptography) which is vulnerable to quantum attacks. A TLS/SSL cipher suite is a formally defined set of cryptographic algorithms used collectively to secure communications during a TLS/SSL session. A Quantum Resistant TLS/SSL suite includes four core components: • Key Encapsulation (ML-KEM) • Authentication (ML-DSA) • Bulk Encryption (AES-128/256) • Message Authentication Code (SHA-512 or Shake-256) Together these algorithms define how a secure channel is established and ensures confidentiality and integrity of data. A Quantum Safe Connectivity Quantum Resistant TLS/SSL Suite with MLKEM and ML-DSA ensures long-term protection of digital communication, shielding data and devices against future quantum threats. It secures internet, cloud and IoT systems that stay safe against future quantum threats.
	Details of Contact Person alongwith Mobile No. and e-mail ID	Mr. Raghavendra S. Patil, C-DAC Bangalore 9845266882 rspatil@cdac.in

Sr. No.	Parameters	Brief details
6.	**Brief details of Technology Developed**	Custom Hardware IP Cores for Post Quantum Cryptography
	Name of the Project under which technology is developed	Development of Secure Post Quantum Public Key Infrastructure
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	4
	Brief Write-up on the Technology Developed	Custom Hardware IP Cores for Post Quantum Cryptography, optimized for speed, memory and security. These IP Cores form the base of nextgeneration secure systems. The following IPCores have been developed 1. AES 128/256 2. ML-KEM 3. ML-DSA 4. SHA3
	Details of Contact Person alongwith Mobile No. and e-mail ID	Mr. Raghavendra S. Patil, C-DAC Bangalore 9845266882 rspatil@cdac.in
7.	**Brief details of Technology Developed**	Development of hybrid quantum classical machine learning frameworks and classical simulations of quantum algorithms for routing, evacuation planning, and hydrological modeling applications.
	Name of the Project under which technology is developed	Quantum Machine Learning Usecases and applications
	Funding Agency	MeitY
	Technology Readiness Level (TRL)	**TRL 3–4** (Analytical and experimental proof-of-concept through simulations and Dashboard for classical flood simulation.
	Brief Write-up on the Technology Developed	It involves the design and evaluation of Quantum Machine Learning (QML) algorithms and hybrid quantum classical approaches for routing and evacuation planning problem. Classical simulations of quantum algorithms such as Quantum Approximate Optimization Algorithm (QAOA), Variational Quantum Circuits (VQC), and Quantum Support Vector Machines (QSVM) have been developed and benchmarked against classical methods. The framework has been applied to routing and evacuation planning scenarios for emergency management, and decision support under extreme conditions. The work establishes a foundation for future toward next generation decision-support systems in disaster management and infrastructure planning.
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